THE TWINS

Gemini

SIZE RANKING 30

BRIGHTEST STAR Pollux (β) 1.2

GENITIVE Geminorum

ABBREVIATION Gem

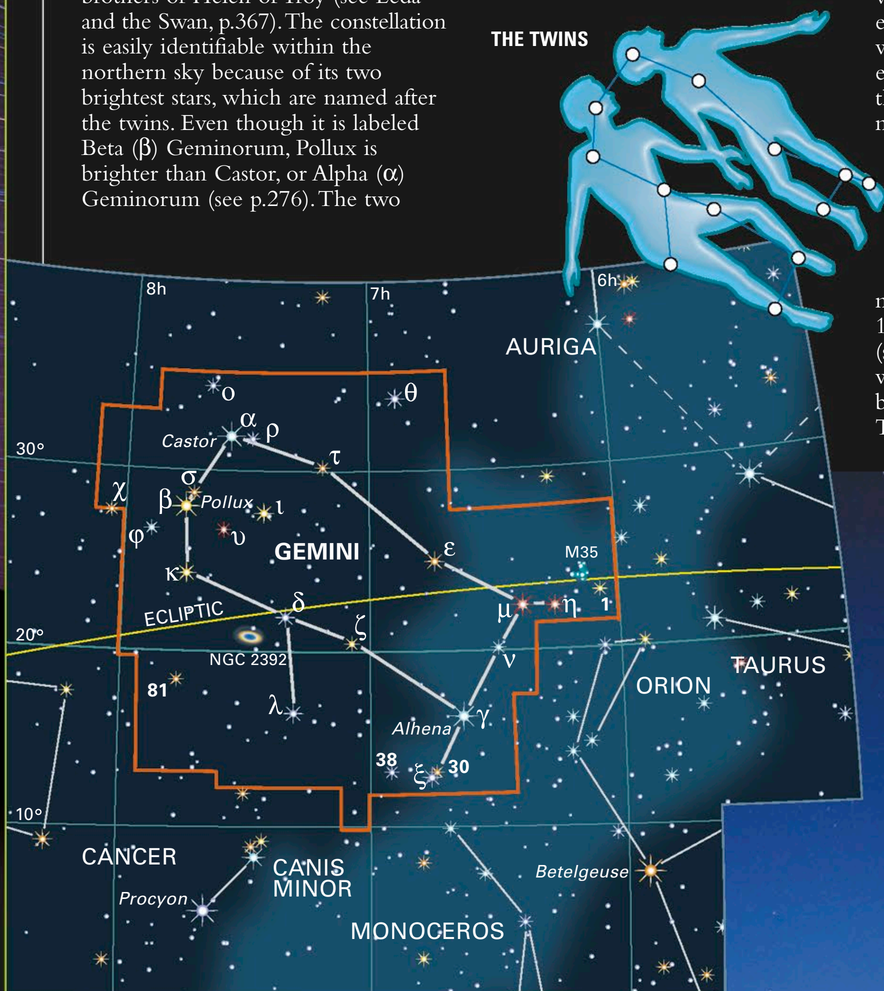
HIGHEST IN SKY AT 10 P.M. January–February

FULLY VISIBLE 90°N–55°S

This prominent zodiacal constellation represents the mythical twins Castor and Pollux, who were the sons of Queen Leda of Sparta and the brothers of Helen of Troy (see Leda and the Swan, p.367). The constellation is easily identifiable within the northern sky because of its two brightest stars, which are named after the twins. Even though it is labeled Beta (β) Geminorum, Pollux is brighter than Castor, or Alpha (α) Geminorum (see p.276). The two stars mark the heads of the twins, while their feet lie bathed in the Milky Way. In mid-December each year, the Geminid meteors radiate from a point in Gemini near Castor.

SPECIFIC FEATURES

Castor is a remarkable multiple star. To the naked eye, it appears as a single entity of magnitude 1.6, but through a small telescope with suitably high magnification, it divides into a sparkling blue-white duo of 2nd and 3rd magnitudes. The two stars form a genuine binary, with an orbital period of 450 years, which also has a 9th-magnitude red dwarf companion. Although these three stars cannot be divided further visually, each is a spectroscopic binary, bringing the total number of stars in the Castor system to six.



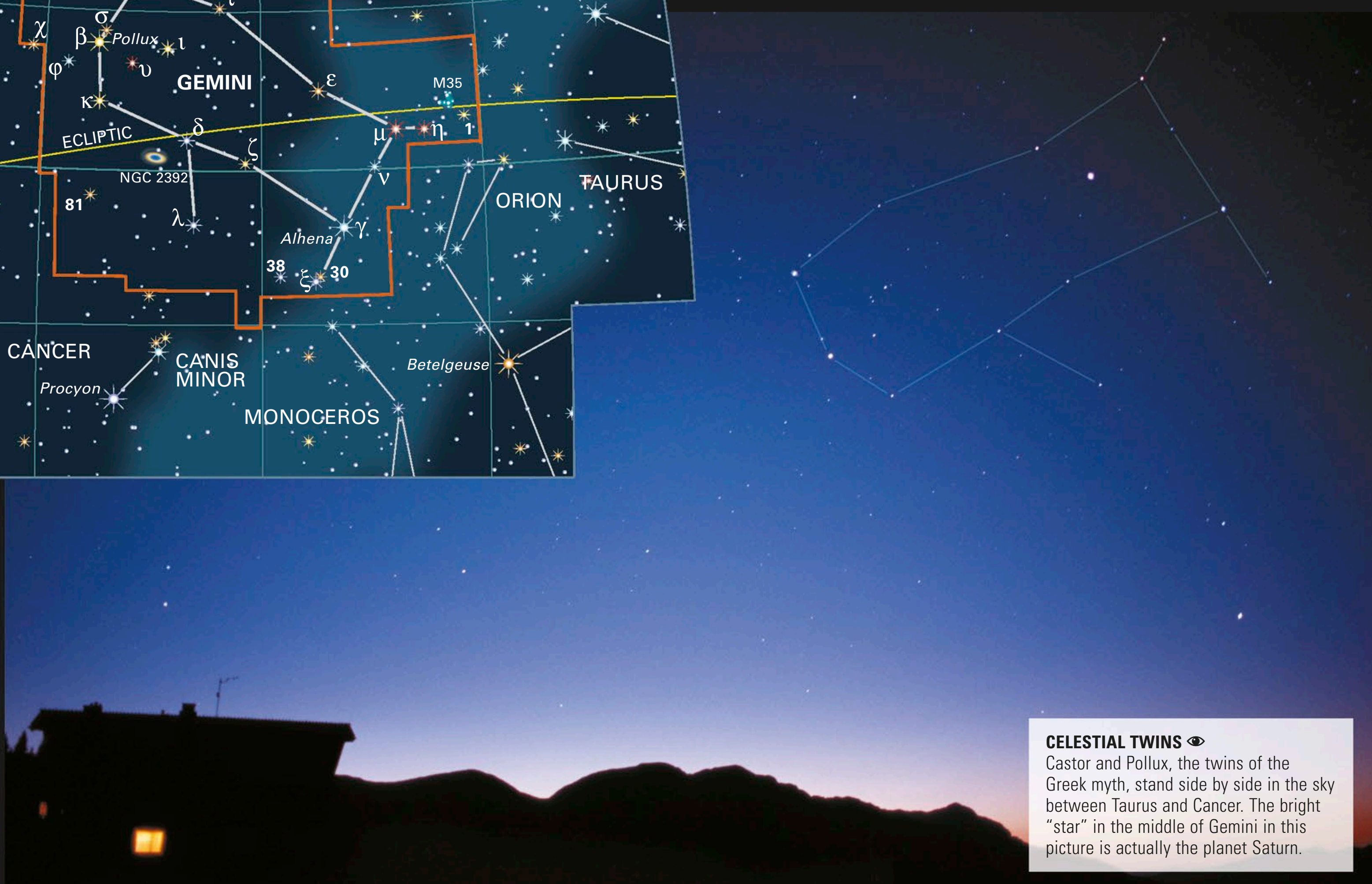
Although Castor and Pollux are named after twins, the stars themselves are far from identical. Being an orange giant, Pollux is noticeably warmer-toned than Castor. It is also closer to Earth, lying only 34 light-years away, compared to Castor's 51 light-years.

The open star cluster M35 lies at the feet of the twins. Under clear skies, this cluster can be glimpsed with the naked eye, but it is more easily found with binoculars, through which it appears as an elongated, elliptical patch of starlight spanning the same apparent width as a full moon. When viewed through a small telescope, its individual stars seem to form chains or curved lines.

Two variable stars of note in Gemini are Zeta (ζ) Geminorum (see p.286), which is a Cepheid variable that ranges between magnitudes 3.6 and 4.2 every 10.2 days, and Eta (η) Geminorum (see p.284), which is a red giant whose brightness can vary anywhere between magnitudes 3.1 and 3.9. This constellation also contains the Eskimo Nebula, or NGC 2392 (see p.259), a planetary nebula with a bluish disk similar in size to that of the globe of Saturn and visible through a small telescope. Larger telescope apertures are needed to reveal the nebula's surrounding fringe of gas, reminiscent of an Inuit parka, that gives NGC 2392 its popular name. An alternative name for this nebula is the Clown-Face Nebula.

THE ESKIMO NEBULA
The planetary nebula NGC 2392 is so called because it is surrounded by a fringe of gas that resembles the fur-lined hood of an Inuit parka.

LARGE AND SMALL CLUSTER
The large star cluster M35 is visible through binoculars; larger telescopes reveal a fainter and more distant cluster, NGC 2158 (bottom right), in the same field of view.



CELESTIAL TWINS
Castor and Pollux, the twins of the Greek myth, stand side by side in the sky between Taurus and Cancer. The bright "star" in the middle of Gemini in this picture is actually the planet Saturn.